

## **Social Media & Network Analytics**

### **Worksheet**

**Do Polarized Emotional Communities Form Around AI Job Replacement Discussions on YouTube, and Who Are the Influential Users Shaping These Communities?**

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**File Name:** Worksheet\_S4186963\_S4190295\_PG\_Group\_39  
**Course:** COSC2671 — Social Media & Network Analytics  
**Level:** Postgraduate (PG)  
**Group:** Group 39  
**Members:** Chantelle Isabelle Dmonte (S4186963)  
Utkarsh Nautiyal (S4190295)

## SECTION 1 — TEAM PROJECT PLAN

**Research Question:** Do polarised emotional communities form around AI job replacement discussions on YouTube and Bluesky?

**Platforms:** YouTube (comments) & Bluesky (posts & replies)

| Week         | Task / Milestone                                                                  | Responsible | Status    |
|--------------|-----------------------------------------------------------------------------------|-------------|-----------|
| <b>Wk 9</b>  | <b>6 -7 May 2026 — Research question finalisation &amp; data collection setup</b> |             |           |
|              | Agree on research question & platform focus (YouTube + Bluesky)                   | Both        | Completed |
|              | Configure YouTube Data API v3 & Bluesky AT Protocol credentials                   | Utkarsh     | Completed |
|              | Design data collection strategy (keywords, date range, volume targets)            | Chantelle   | Completed |
|              | Initial literature review on polarisation metrics                                 | Chantelle   | Completed |
| <b>Wk 9</b>  | <b>8-10 May 2026 — Data collection &amp; preprocessing</b>                        |             |           |
|              | Collect YouTube comments & Bluesky posts (target $\geq 1,000$ each)               | Utkarsh     | Completed |
|              | Text preprocessing pipeline (tokenisation, stopword removal, lemmatisation)       | Utkarsh     | Completed |
|              | VADER sentiment analysis on both datasets                                         | Chantelle   | Completed |
|              | Dataset overview statistics & initial EDA                                         | Utkarsh     | Completed |
| <b>Wk 10</b> | <b>11-17 May 2026 — Network construction &amp; NLP analysis</b>                   |             |           |
|              | Build reply-graph networks (YouTube & Bluesky)                                    | Chantelle   | Completed |
|              | Compute centrality measures (PageRank, betweenness, in-degree)                    | Chantelle   | Completed |
|              | LDA topic modelling with perplexity validation                                    | Utkarsh     | Completed |
|              | Community detection via Louvain algorithm                                         | Both        | Completed |
| <b>Wk 11</b> | <b>18-24 May 2026 — Analysis, visualisation &amp; report writing</b>              |             |           |
|              | Community sentiment analysis & polarisation index                                 | Chantelle   | Completed |
|              | Statistical tests (Kruskal-Wallis, Mann-Whitney)                                  | Chantelle   | Completed |
|              | Network visualisations (centrality scatter, community bar charts)                 | Utkarsh     | Completed |
|              | Cross-platform comparison heatmap                                                 | Chantelle   | Completed |

|              |                                                               |           |           |
|--------------|---------------------------------------------------------------|-----------|-----------|
|              | Draft report — Sections 1–3 (Introduction, Data, Methods)     | Both      | Completed |
| <b>Wk 12</b> | <b>25 May 2026 — Final report, submission &amp; worksheet</b> |           |           |
|              | Complete Sections 4–5 (Results & Discussion)                  | Chantelle | Completed |
|              | Finalise worksheet (timesheets + self-reflections)            | Both      | Completed |
|              | Code cleanup, GitHub submission                               | Both      | Completed |

| Week | Task / Milestone                | Responsible | Status      |
|------|---------------------------------|-------------|-------------|
|      | Final submission                | Both        | Completed   |
|      | Slide creation for presentation | Both        | In progress |

## SECTION 2 — WEEKLY TIMESHEETS

### Timesheet — CHANTELLE ISABELLE DMONTE (S4186963)

| Week                             | Dates | Tasks Completed                                                                                                                | Mon–Wed (hrs) | Thu–Fri (hrs) | Total Hrs |
|----------------------------------|-------|--------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-----------|
| <b>Week 9</b><br>6–10 May 2026   |       | Research question finalisation<br>API credential setup<br>Data collection strategy design<br>Literature review on polarisation | 12            | 5             | 17        |
| <b>Week 10</b><br>11–17 May 2026 |       | Data collection (YouTube + Bluesky)<br>Text preprocessing pipeline<br>VADER sentiment analysis<br>EDA & dataset overview       | 10            | 3             | 13        |
| <b>Week 11</b><br>18–24 May 2026 |       | Reply-graph construction<br>network Centrality measures computation<br>LDA topic modelling<br>Community detection (Louvain)    | 15            | 5             | 20        |

|                                              |                                                                                                                                                                                                           |    |   |    |
|----------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----|---|----|
| <p><b>Week 12</b><br/>25 May–30 May 2026</p> | <p>Community sentiment analysis<br/>Statistical tests<br/>Visualisations &amp; charts<br/>Draft report sections<br/>Final report completion<br/>Peer review<br/>Worksheet finalisation<br/>Submission</p> | 10 | 5 | 15 |
| TOTAL HOURS                                  |                                                                                                                                                                                                           |    |   | 65 |

## Timesheet — UTKARSH NAUTIYAL (S4190295)

| Week                                 | Dates | Tasks Completed                                                                                                                                                                         | Mon–Wed (hrs) | Thu–Fri (hrs) | Total Hrs |
|--------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|---------------|-----------|
| <b>Week 9</b><br>6–10 May 2026       |       | Research question finalisation<br>API credential setup<br>Data collection strategy design<br>Literature review on polarisation                                                          | 10            | 5             | 15        |
| <b>Week 10</b><br>11–17 May 2026     |       | Data collection (YouTube + Bluesky)<br>Text preprocessing pipeline<br>VADER sentiment analysis<br>EDA & dataset overview                                                                | 8             | 4             | 12        |
| <b>Week 11</b><br>18–24 May 2026     |       | Reply-graph construction<br>network Centrality measures computation<br>LDA topic modelling<br>Community detection (Louvain)                                                             | 10            | 5             | 15        |
| <b>Week 12</b><br>25 May–30 May 2026 |       | Community sentiment analysis<br>Statistical tests<br>Visualisations & charts<br>Draft report sections<br>Final report completion<br>Peer review<br>Worksheet finalisation<br>Submission | 10            | 5             | 15        |
| <b>TOTAL HOURS</b>                   |       |                                                                                                                                                                                         |               |               | <b>57</b> |

## SECTION 3 — INDIVIDUAL SELF-REFLECTION

### Self-Reflection — CHANTELLE ISABELLE DMONTE (S4186963)

Working on this project gave me a meaningful opportunity to apply social media analytics techniques to a real-world research question about AI discourse and community polarisation. My primary contributions spanned data collection pipeline design, preprocessing, VADER sentiment analysis, and LDA topic modelling with perplexity-based validation — all implemented in Python using a checkpoint system to preserve YouTube API quota across runs.

One of the key technical challenges I encountered was the ATS-clean data ingestion issue with Bluesky's `reply_to_handle` field: the raw AT Protocol URI format required a custom `extract_reply_target()` function before a valid reply graph could be constructed. Debugging this iteratively — printing diagnostics, inspecting dtypes, and patching the graph builder call — reinforced my appreciation for defensive data engineering practices.

Collaborating with Utkarsh helped me improve my approach to task decomposition and asynchronous code review. In hindsight, I would invest more time upfront in agreeing on a shared data schema before each of us began independent collection modules, which would have reduced integration friction in Week 11.

Overall, this project deepened my understanding of network-based approaches to studying online discourse, and strengthened my practical skills in graph analytics and NLP pipelines — areas directly relevant to my Master of AI studies and future graduate roles in data/AI engineering.

### **Self-Reflection — UTKARSH NAUTIYAL (S4190295)**

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This project provided an excellent opportunity to bridge theoretical knowledge of network analysis with hands-on implementation across two distinct social media platforms. My primary contributions included network construction (building and validating both the YouTube and Bluesky reply graphs), centrality computation, Louvain community detection, and the cross-platform comparison visualisations.

A significant challenge was ensuring methodological consistency between the two platforms despite their structural differences YouTube's nested comment threads versus Bluesky's decentralised AT Protocol reply chains. Adapting the graph builder to handle both formats without losing directed edge information required careful review of each platform's data model.

Working with Chantelle was productive and well-structured; her systematic approach to preprocessing and sentiment analysis made the downstream network tasks more reliable. One area for improvement in future projects would be establishing shared integration tests early, so that each module's outputs are validated against expected schemas before being passed to the next stage.

This experience has consolidated my skills in NetworkX, Python-Louvain, and statistical testing (Kruskal-Wallis), and has given me a richer understanding of how online communities organise around emotionally charged topics insight I look forward to building on in my research and professional career.